Magnetostratigraphy of the uppermost Triassic and lowermost Jurassic Moenave Formation, western United States: Correlation with strata in the United Kingdom, Morocco, Turkey, Italy, and eastern United States

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ABSTRACT

A composite magnetostratigraphy based on the magnetic polarity data from four sections of the uppermost Triassic and lowermost Jurassic Moenave Formation, Utah and Arizona, USA, can be correlated to the marine successions at Saint Audrie's Bay (UK), Oyuklu, Turkey, and the Southern Alps, Italy, and to the nonmarine sections in Morocco, northern Africa, and the Newark Basin, eastern North America, all deposited across the Triassic–Jurassic boundary. Our proposed correlation provides a stratigraphic framework to tie Triassic–Jurassic sedimentation in the American Southwest to the marine UK, Turkey, and Italy sections, and to the Pangea rift history, including extrusive igneous rocks, preserved in Morocco and in the Newark Basin. The Moenave polarity record is characterized by mostly normal polarity, as is consistent with other polarity records across the Triassic–Jurassic boundary, and is interrupted by at least two well-defined reverse-polarity magnetozones. On the basis of available paleontologic information, we interpret the oldest well-defined, reverse-polarity magnetozone, M2r of the Moenave Formation, to correlate with SA5n.2r or SA5n.3r of the Saint Audrie's Bay record, H– of the Oyuklu record, BIT5n.1r of the Italcementi Quarry record, the oldest reverse magnetozone in sedimentary rocks in Morocco, and with reverse magnetozone E23r of the Newark Basin. The youngest reverse magnetozone of the Moenave Formation, M3r, is correlated to the latest Triassic magnetozones SA5n.5r of the Saint Audrie's Bay record, J– of the Oyuklu record, and with the interval of reverse polarity in the "intermediate unit" of the Morocco record. Magnetostratigraphic correlations and marine biostratigraphic information support placement of the Triassic–Jurassic boundary in the middle to upper Whitmore Point Member of the Moenave Formation, the Lias Group of the Saint Audrie's Bay section, the chert-rich limestone of the Oyuklu section, above the Zu Limestone in Italy, and in the central Atlantic magmatic province extrusive zone in the Morocco and the Newark records.

INTRODUCTION

The interval of time represented by the transition from the Late Triassic to the Early Jurassic is an important one in Earth history, because it is associated with substantial faunal and floral evolution, including major marine and terrestrial extinctions and voluminous magmatism related to the opening of the Atlantic Ocean (Hesselbo et al., 2002; Knight et al., 2003; Olsen et al., 2003; Marzoli et al., 2004; Tanner et al., 2004; Hesselbo et al., 2007; Kozur and Weems, 2007; Lucas and Tanner, 2007b; Palfy et al., 2007; Tanner and Lucas, 2007; Whiteside et al., 2007; Tanner, 2008; Cirilli et al., 2009; Lucas and Muttoni et al., 2009; Deenen et al., 2010; Schoene et al., 2010). Our understanding of the temporal relationships of events near the Triassic–Jurassic (T-J) boundary, such as faunal extinction and the global effects of extrusion of the Pangean central Atlantic magmatic province, is limited because there is no single biostratigraphic marker to define the T-J boundary in both marine and terrestrial records.

Biostratigraphic records in the American Southwest include footprints, conchostracans, and palynomorphs and suggest the T-J boundary lies within the Moenave Formation of the Glen Canyon Group (Lucas and Heckert, 2001; Cornet and Waanders, 2006; Lucas and Tanner, 2007a; Tanner and Lucas, 2007; Tanner and Lucas, 2009a; Kozur and Weems, 2010). In the UK, the Triassic–Jurassic section at Saint Audrie's Bay (Hounslow et al., 2004) is of particular importance, because it provides the most complete magnetostratigraphic record in marine strata associated with the long-used marine biostratigraphic definition of the T-J boundary, the lowest occurrence of the Jurassic ammonite *Psiloceras planorbis*. However, the T-J boundary at Saint Audrie's Bay may occur within a 6-m-long interval between the highest occurrence of conodonts and the lowest occurrence of the ammonite *P. planorbis* (Hounslow et al., 2004). In Oyuklu, Turkey, the T-J boundary is identified within the chert-rich limestone by the highest occurrence of conodonts. At the Italcementi Quarry section in the Southern Alps, Italy, which includes upper Rhaetian and lowermost Hettangian shallow-marine carbonate rocks, Muttoni et al. (2009) associate the T-J boundary interval with the acme of *Kraeuselisporites reissingeri*, which is within the interval of a negative carbon isotope excursion. The record obtained from the High Atlas section in Morocco (Knight et al., 2003; Marzoli et al., 2004; Knight et al., 2004; Deenen et al., 2010) contains robust magnetic polarity data and isotopic age determinations associated with the central Atlantic magmatic province eruptions, but it has not provided biostratigraphic information. The T-J boundary in the Moroccan record lies within the "upper unit." In eastern North America, Newark Supergroup strata and intercalated central Atlantic magmatic province extrusive rocks in the Newark Basin also span the T-J boundary, as indicated by biostratigraphic and isotopic age data from the central Atlantic magmatic province extrusive zone (Cornet and Olsen, 1985; Whiteside et al., 2007). In addition, Newark Supergroup strata provide the thickest known stratigraphic section across the boundary (Kent and Olsen, 1999, 2008). An initial placement of the T-J boundary and correlation among all

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of these sections can be obtained by examining the biostratigraphic data and isotopic age estimates (Fig. 1). However, inconsistencies in chronostratigraphic data among these sections exist and hinder our ability to precisely locate and correlate the T-J boundary.

Several criteria have been proposed for defining the T-J boundary, including marine events such as the highest occurrence of conodonts, the lowest occurrence of psiloceratid ammonites, terrestrial events such as pollen turnover (the end-Triassic event), and crurotarsan extinction near the boundary (Lucas and Tanner, 2007a, 2007b). The relative synchrony of each proposed definition cannot be assessed without an independent chronostratigraphic framework. Magnetostratigraphy is a powerful method for establishing an independent record of geologic time (Opdyke and Channell, 1996). Polarity records of the geomagnetic field are global and independent of processes such as widespread magmatism, extinction, or climate change. Nevertheless, magnetostratigraphic records may be compromised, especially in fluvial and lacustrine environments, due to hiatuses, variable sedimentation rates, and the ability of sediments to accurately record the geomagnetic field. Therefore, it is important to study multiple sections to establish consistency in polarity sequences, to acquire a high-resolution sampling, to assess the timing of characteristic remanence magnetization acquisition, to perform field and laboratory tests for remanence stability, and to fully integrate magnetostratigraphic interpretations with biostratigraphic observations.

We hypothesize that, with sufficiently detailed sampling, the Moenave Formation polarity record can be better defined and more completely correlated to the Saint Audrie's Bay, Oyuklu, Southern Alps of Italy, High Atlas, and Newark polarity records. To test our hypothesis, we obtained paleomagnetic data from four previously unexamined, well-exposed sections of Moenave Formation strata in southwestern Utah and northern Arizona (Fig. 2). The first paleomagnetic study of the Moenave Formation by Ekstrand and Butler (1989), in the Vermillion Cliffs of northern Arizona, indicated that the Moenave Formation is dominated by normal polarity. A subsequent study by Molina-Garza et al. (2003) in the nearby Echo Cliffs of northern Arizona supported the conclusion that these rocks yield mostly normal-polarity magnetizations. Both studies, however, reported one thin, reverse-polarity magnetozone near the top of the Moenave Formation. We report additional short, reverse-polarity magnetozones in both the Dinosaur Canyon and Whitmore Point members of the Moenave Formation and use these magnetozones to make correlations with coeval strata in UK, Turkey, Italy, Morocco, and eastern North America.

THE TRIASSIC–JURASSIC BOUNDARY

In the absence of a global stratotype for the base of the Jurassic period, several biostratigraphic events including, but not limited to, crurotarsan ("thecodont") extinction, changes in terrestrial pollen, and the appearance of smooth-shelled psiloceratid ammonites, have been used to identify the T-J boundary (Gradstein et al., 2004; Hounslow et al., 2004; Kent and Olsen, 1999, 2008; Lucas and Tanner, 2007b; Whiteside et al., 2007; Lucas, 2009; van de Schootbrugge et al., 2009). This array of biostratigraphic indicators has resulted in an interval over which the T-J boundary could be defined. This interval could be as thin as 6 m in marine strata, ~10 m on the Colorado Plateau, and possibly up to several hundred meters in the Newark rift-basin strata. A distinctive carbon isotope excursion has also been identified in this T-J boundary interval (Palfy et al., 2001; Hesselbo et al., 2002; Kuerschner et al., 2007; Palfy et al., 2007; Muttoni et al., 2009). This $\delta^{13}C$ isotopic record has an initial negative excursion, on the order of ~4‰, in the latest Triassic, a slight recovery, and a main negative

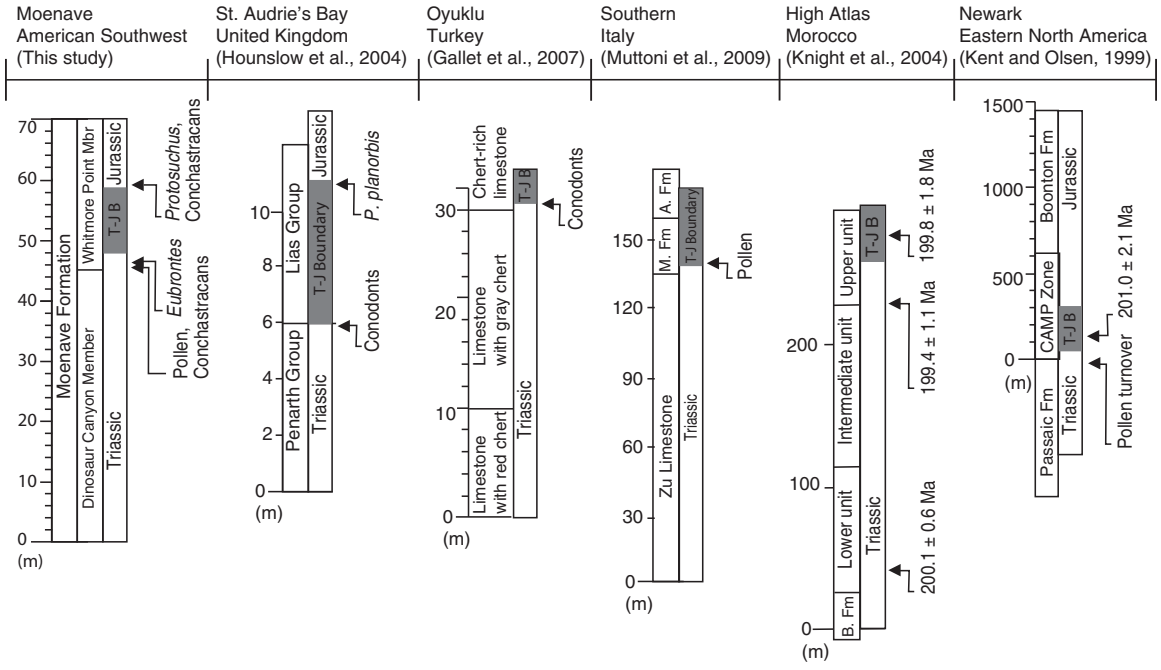


Figure 1. Stratigraphic sections of the Moenave Formation, American Southwest, Saint Audrie's Bay, UK, Oyuklu, Turkey, Italcementi Quarry (M. Fm.—Malanotte Formation; A. Fm.—Albenza Formation), Italy, High Atlas (B. Fm.—Bigoudine Formation), Morocco, and Newark Basin of eastern North America. Biostratigraphic data and isotopic age estimates are given for each section. Vertical scale is in meters. CAMP—central Atlantic magmatic province.

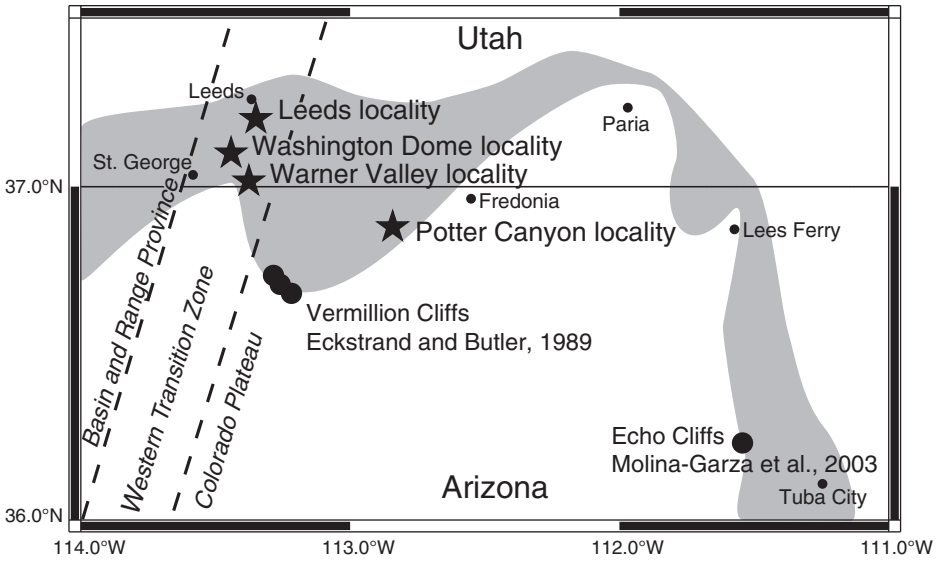


Figure 2. The shaded region represents the northern and southern Moenave outcrop belts. Index map of Moenave localities sampled shown by stars: Leeds at 37.202° N, 246.660° E; Washington Dome at 37.009° N, 246.558° E; Warner Valley at 37.016° N, 246.625° E; Potter Canyon at 36.879° N, 247.156° E. Small circles show nearby populated areas. Large circles represent previous sampling localities reported in Ekstrand and Butler (1989, Vermillion Cliffs) and Molina-Garza et al. (2003, Echo Cliffs).

excursion that begins near the T-J boundary and continues into the Jurassic. The isotopic excursion has been interpreted to reflect episodic central Atlantic magmatic province volcanism associated with the opening of the Atlantic Ocean. A U/Pb zircon isotopic age determination provides an age estimate for Newark central Atlantic magmatic province flows near the T-J boundary of ~201.0 ± 2 Ma, and a $^{40}Ar/^{39}Ar$ age determination of 199.8 ± 1.8 Ma has been obtained on a central Atlantic magmatic province flow in Morocco (Kent and Olsen, 1999; Knight et al., 2003; Olsen et al., 2003; Knight et al., 2004). A more exact resolution of the absolute age of the T-J boundary is hindered by differences in age estimates calculated in different isotopic systems (Lucas and Tanner, 2007b) and the precision of the methods.

The working definition of the T-J boundary has long been the lowest stratigraphic occurrence of the psiloceratid ammonite *Psiloceras planorbis* (Gradstein et al., 2004; Lucas and Tanner, 2007b) defined in the marine Saint Audrie's Bay section. The youngest definitive Triassic biostratigraphic marker in the Saint Audrie's Bay record is the highest occurrence of conodonts. Therefore, the T-J boundary in the southern UK record must lie in the ~6-m-thick interval between the highest occurrence of conodonts and the lowest occurrence of *P. planorbis* (Fig. 1). The main $\delta^{13}C$ isotopic excursion also lies in this 6 m interval (Hesselbo et al., 2002).

Recently, the Subcommission on Jurassic Stratigraphy of the International Commission on Stratigraphy (ICS) proposed placing the base of the Jurassic slightly lower, at the lowest stratigraphic occurrence of the ammonite *Psiloceras spelae*, based on a stratotype at Kuhjoch, Austria. This definition of the base of the Jurassic is not particularly useful in this study because there is no magnetostratigraphic record for the Kuhjoch section, and the ammonite *P. spelae* is not present in any of the records considered here. However, we adhere to using *P. spelae* as the base of the Jurassic and note that *P. spelae* is stratigraphically lower than *P. planorbis* but higher than other biomarkers used to interpret the T-J boundary.

In Oyuklu, Turkey, conodonts are the most prevalent biostratigraphic marker (Gallet et al., 2007). The highest occurrence of conodonts is at the top of the limestone with gray chert interval ~30 m above the base of the sections (Fig. 1). The T-J boundary must lie stratigraphically above this interval. In the Italcementi Quarry section of the Southern Alps, the Triassic–Jurassic boundary is inferred to occur at the disconformity between the Zu Limestone, which exposes the highest occurrence of conodonts, and the Malanotte Formation, which contains the lowest occurrence of Hettangian pollen (i.e., *Classopollis torosus*, *Gliscopollis meyeriana*, *Microreticulatisporites fuscus*, *Callialasporites spp.*, *Retitriletes austroclavatidites*, *R. clavatoides*, *Tsugapollenites pseudomassulae*, and rare occurrences of *Cerebropollenites macroverrucosus*) (Muttoni et al., 2009). However, the stratigraphic boundary between the Zu Limestone and the Malanotte Formation predates the lowest occurrence of the ammonite *P. spelae* (Muttoni et al., 2009) and is therefore older than the Triassic–Jurassic boundary.

In the nonmarine Newark Supergroup strata of eastern North America, the T-J boundary lies in the sequence of rocks deposited after the onset of central Atlantic magmatic province volcanism (Lucas and Tanner, 2007b; Lucas, 2009; Kozur and Weems, 2010). The Newark pollen event, preserved in strata of the Passaic Formation below the central Atlantic magmatic province extrusive zone in the Newark Basin, was previously used to estimate the T-J boundary (Cornet and Olsen, 1985; Fowell and Olsen, 1993; Olsen et al., 2003). The upper Passaic Formation, Newark Basin, contains *Eubrontes* dinosaur tracks similar to tracks of Norian or Rhaetian age found in European sections (Litwin, 1991; Hounslow et al., 2004; Lucas and Tanner, 2007b; Whiteside et al., 2007; Lucas, 2009). A reexamination of the palynostratigraphy and the new data provided by conchostracan biostratigraphy (Kozur and Weems, 2010) both lend further support to a latest Triassic age for the upper strata of the Passaic Formation, and indicate that the T-J boundary is within the central Atlantic magmatic province extrusive zone. Therefore, the Newark pollen event must be considered to be in uppermost Triassic strata (Fig. 1).

In the nonmarine High Atlas record of Morocco, geochemical and magnetostratigraphic data have been used to support placement of the T-J boundary within the "upper unit" of the central Atlantic magmatic province flows (Marzoli et al., 1999; Knight et al., 2004). A new study by Deenen et al. (2010) on the red bed sedimentary sequence from the Bigoudine Formation in the Agrana Basin, Morocco, uses astrochronology, magnetostratigraphy, basalt geochemistry, and geobiology to establish a correlation between an interval of reverse polarity identified in the upper Bigoudine Formation and the short reverse-polarity magnetozone E23r in the upper Passaic Formation in the Newark Basin. Previous work in the Moroccan section identified two intervals of reverse polarity within the "intermediate unit" basalt flow sequence (Knight et al., 2004). However, because each reverse-polarity magnetozone was identified at completely separate sections, it is possible that the two intervals of reverse polarity are actually the same event (Knight et al., 2004; Marzoli et al., 2004; Deenen et al., 2010). Isotopic age estimates ($^{40}Ar/^{39}Ar$ step-heating analyses) on the High Atlas, Moroccan basalt flows provide an

age estimate of 199.8 ± 1.8 Ma for the "upper unit" (Fig. 1). Recently obtained paleomagnetic data from the North Mountain Basalt, the record of initial central Atlantic magmatic province magmatism in the Fundy Basin of Nova Scotia, have been interpreted by Kent et al. (2009) to record the same geomagnetic field behavior as the Lower and Intermediate basalt sequences of the Morocco section, as reported by Knight et al. (2004). For the North Mountain Basalt, Kent et al. (2009) reported no evidence of the field excursions that have been identified in the "intermediate unit" of the Moroccan section.

As discussed above, the biostratigraphic record for strata of the northern outcrop belt of the Moenave Formation suggests that the T-J boundary is within the Whitmore Point Member of the Moenave Formation. Thus, the T-J boundary must lie within the middle to upper strata of the Whitmore Point Member between unambiguously Triassic and Jurassic fossil-bearing strata (Fig. 1). Identification of the T-J boundary in all six records (Saint Audrie's Bay, Turkey, Italy, Morocco, Newark Basin, and the American Southwest) as well as its correlation among these regions, is complicated by the fact that a normal-polarity magnetozone of long duration includes all the local T-J boundary indicators, making accurate correlation of any short-lived reverse-polarity magnetozones in these sections of paramount importance (Molina-Garza et al., 2003; Hounslow et al., 2004; Knight et al., 2004; Gallet et al., 2007; Kent and Olsen, 2008; Muttoni et al., 2009). Short-lived reverse-polarity magnetozones near the Triassic–Jurassic boundary have been identified at each Moenave locality. The purpose of this work is to establish a more robust polarity record for the Moenave Formation and to propose a correlation of the reverse magnetozones near the T-J boundary in western North America, southern UK, southwest Turkey, Southern Alps, Italy, northwestern Africa, and eastern North America sections deposited across the Triassic–Jurassic boundary.

GEOLOGY

Harshbarger et al. (1957) named the Moenave Formation as the basal formation of the Glen Canyon Group. The Moenave Formation includes two members, the Dinosaur Canyon and Whitmore Point (Lucas and Tanner, 2006; Tanner and Lucas, 2007; Tanner and Lucas, 2009b). The basal Dinosaur Canyon Member is separated from the underlying Upper Triassic strata of the Owl Rock Formation of the Chinle Group by an unconformity that decades ago was termed the J-0 unconformity, when it was believed to coincide with the T-J boundary (Pipiringos and O'Sullivan, 1978), but is better referred to as the Tr-5 unconformity, a within-Triassic hiatus (Lucas and Tanner, 2006; Tanner and Lucas, 2007). The Whitmore Point Member conformably overlies Dinosaur Canyon strata. The Springdale Sandstone Member of the Kayenta Formation disconformably overlies the upper Whitmore Point strata at the sub-Springdale unconformity (Tanner and Lucas, 2007). The Moenave Formation is exposed in a northern outcrop belt, between southern Utah and northernmost Arizona, and a southern outcrop belt, in north-central Arizona (Fig. 2).

Exposures of the Moenave Formation in the southern outcrop belt (north-central Arizona) are dominated by intermixed eolian and fluvial facies of the Dinosaur Canyon Member (Harshbarger et al., 1957; Tanner and Lucas, 2007). The northern outcrop belt (southern Utah to northern Arizona) contains the basal Dinosaur Canyon Member and the upper, mostly lacustrine facies Whitmore Point Member (Tanner and Lucas, 2009b). Strata of the Whitmore Point Member are not preserved in the southern outcrop belt. These lacustrine strata were deposited in one large lake ("Lake Dixie") or in a mosaic of smaller lakes in the area between Saint George and Kanab, Utah (Kirkland et al., 2002; Milner et al., 2006; Tanner and Lucas, 2009b).

Abundant dinosaur tracks assigned to the ichnogenera *Eubrontes* and *Grallator* are preserved in lower Whitmore Point strata at the Dinosaur Discovery Site at Johnson Farm in Saint George, Utah, near the Washington Dome locality (Kirkland et al., 2002; Milner et al., 2006). The lowermost strata of the Whitmore Point Member also yield palynomorphs, conchostracans (*Euestheria* and *Bulbilimnadia*), collected at the Potter Canyon locality by previous workers, and tetrapod tracks (*Eubrontes*) consistent with a latest Triassic age (Cornet and Waanders, 2006; Kozur and Weems, 2010). Upper Whitmore Point strata yield Jurassic conchostracans, such as *Bulbilimnadia kilianorum*. Strata of the uppermost Dinosaur Canyon Member in the southern outcrop belt (not sampled in this study) that are laterally equivalent to the upper Whitmore Point strata in the northern outcrop belt contain tetrapod body fossils (especially the crocodylomorph *Protoscuhus*) indicative of an Early Jurassic age (Lucas and Heckert, 2001; Lucas, 2009). Previously published biostratigraphic evidence thus suggests that the T-J boundary is located in the Whitmore Point Member.

METHODS

Moenave Formation localities in the northern outcrop belt, on and off the Colorado Plateau, were chosen because of stratal continuity with underlying and overlying formations and essentially complete exposure of both the fluvial facies of the Dinosaur Canyon and lacustrine facies of the Whitmore Point members. Four localities in Moenave Formation strata were sampled between Saint George, Utah, and Pipe Springs, Arizona: Leeds (average strike/dip [s/d]: 50/30 SE), Warner Valley (s/d: 62/36 SE), Washington Dome (s/d: 227/12 NW), and Potter Canyon (s/d: 256/6 NW) (Fig. 2). The Leeds, Warner Valley, and Washington Dome localities are within the western transition zone of the Colorado Plateau. The average thickness of the Dinosaur Canyon Member is ~45 m. Average thickness of the Whitmore Point Member is ~20 m. The fourth locality (Potter Canyon) is on the Colorado Plateau and consists of ~10 m of Dinosaur Canyon strata and ~40 m of Whitmore Point strata.

Competent beds in Moenave Formation strata were sampled using a standard gas-powered drill. Fine-grained, less competent intervals, including hematitic mudstones and claystones, were block sampled. One site equals one stratigraphic horizon sampled as seven to ten independently oriented core samples or three to seven blocks. Core specimens (2.2 × 2.5 cm right cylinders) and cubes prepared from oriented blocks (typically by dry sawing) were demagnetized with chemical, alternating field (AF), and thermal methods. Specimens were prepared for chemical demagnetization by cutting slits into the sides of core, thus increasing the surface area (Henry, 1979). These high surface-area specimens were sequentially immersed in 100% reagent grade HCl for 1 min, 3 min, 10 min, 60 min, 12 h, and 24 h (Fig. 3B). The AF demagnetization was carried out to 90 mT. Thermal demagnetization was conducted at 100 °C steps from natural remanent magnetization (NRM) to 400 °C, followed by 25 °C to 50 °C steps to 630 °C, and then 10 °C to 15 °C steps to 685 °C (Figs. 3 and 4). Specimens were also treated with a combination of AF, up to 90 mT, and thermal methods (Fig. 4B). Remanence measurements were made on a three-axis, DC SQUID 2-G Enterprises 760R magnetometer at the University of New Mexico, and 13 sites were measured in the paleomagnetism laboratory at Occidental College. Characteristic remanent magnetization (ChRM) vector components were estimated using principal component analysis (Kirschvink, 1980). Most often, demagnetization vectors trend to the origin, so lines were estimated on anchored data points between ~600 °C and 685 °C. Sixty-nine, out of 533, measured ChRM directions were rejected from site-mean calculations because mean angular deviation values exceeded 15° or the ChRM directions deviated from directions from other samples at that site and thus were not consistent with expected directions for the latest

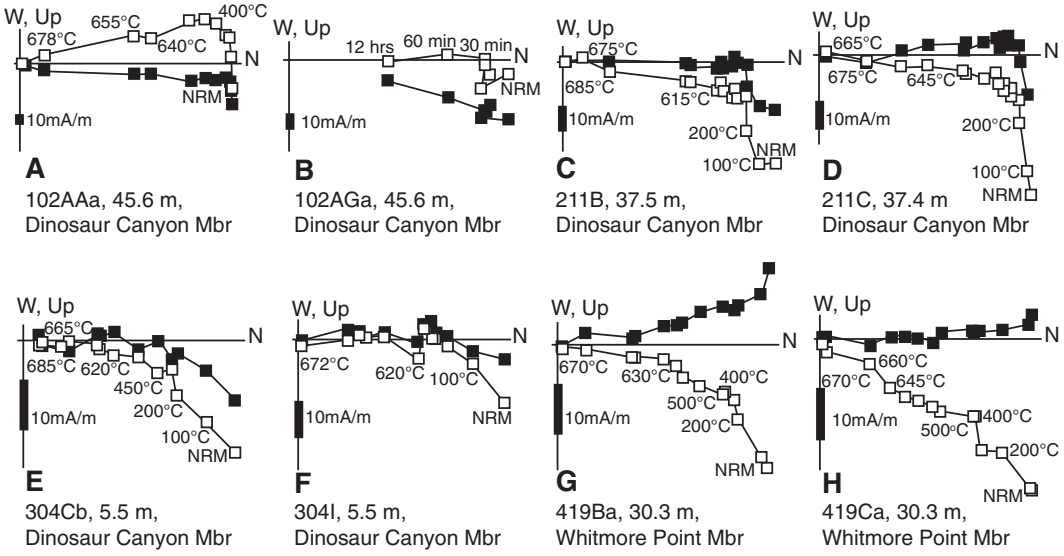


Figure 3. Paleomagnetic data from the Moenave Formation. Orthogonal vector plots show typical characteristic remanent magnetization directions interpreted as normal polarity. Solid squares represent projections on the horizontal plane; open squares represent projection onto a vertical plane. (A) Thermal demagnetization of specimen from the Leeds locality; (B) chemical demagnetization of specimen from the Leeds locality; (C) and (D) thermal demagnetization of specimens from the Washington Dome locality; (E) and (F) thermal demagnetization of specimens from the Warner Valley locality; (G) and (H) thermal demagnetization of specimens from the Potter Canyon locality. Data given in tilt-corrected coordinates. See Table 1 for statistical information. Mbr—Member; NRM—natural remanent magnetization.

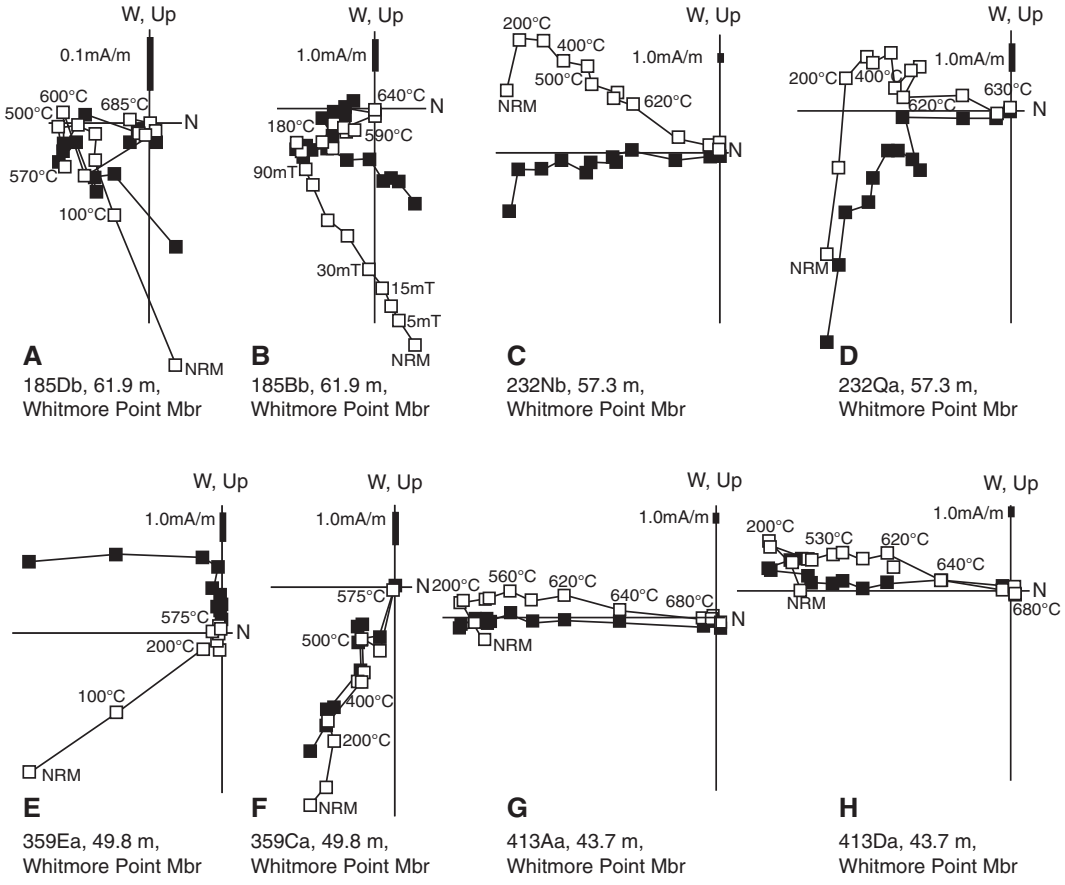


Figure 4. Paleomagnetic data from the Moenave Formation. Orthogonal vector plots show typical characteristic remanent magnetization directions interpreted as reverse polarity. Solid squares represent projections on the horizontal plane; open squares represent projection onto a vertical plane. (A) Thermal demagnetization of specimen from the Leeds locality; (B) combined alternating field and thermal demagnetization of specimen from the Leeds locality; (C) and (D) thermal demagnetization of specimens from the Washington Dome locality; (E) and (F) thermal demagnetization of specimens from the Warner Valley locality; (G) and (H) thermal demagnetization of specimens from the Potter Canyon locality. Data given in tilt-corrected coordinates. See Table 1 for site-mean directions and related statistics. Mbr—Member; NRM—natural remanent magnetization.

Triassic–earliest Jurassic time period for North America. Site-mean directions for 89 individual sites were estimated using Fisher statistics (Fisher, 1953) from linear data on 464 ChRM directions (Fig. 5). Grand mean directions, calculated from those ChRM directions for both normal and reverse polarity, are consistent with expected directions for T-J aged rocks (Fig. 6) (Ekstrand and Butler, 1989; Van der Voo, 1993; Besse and Courtillot, 2002; Molina-Garza et al., 2003; Torsvik et al., 2008). Virtual geomagnetic poles were determined for 79 (out of 89) site-mean directions that have cones of confidence less than or equal to 15° and precision parameters greater than 10 (Table 1).

The magnetic mineralogy of the Moenave Formation samples was investigated using three-component thermal demagnetization of isothermal remanent magnetization (IRM), IRM acquisition and backfield demagnetization, and chemical leaching followed by IRM acquisition. Samples were exposed to decreasing DC magnetic fields of 3 T, 0.3 T, and 0.03 T along orthogonal axes following the method of Lowrie (1990) (Fig. 7). Petrographic analysis in transmitted and reflected light was also carried out on polished thin sections of natural and chemically leached Moenave samples. As discussed below, thin sections in both transmitted and reflected light show hematite as both authigenic and detrital phases (Fig. 8).

A polarity record was compiled for each stratigraphic section by interpreting horizons yielding high northern (positive), virtual geomagnetic pole (VGP) latitudes to be of normal polarity and those yielding high southern (negative) VGP latitudes to be of reverse polarity (Table 1). The reverse-polarity magnetozones identified in the Whitmore Point Member were used as tie points for correlation between the four localities, because magnetozones represent time-equivalent strata. The thickness of each member was averaged to build a composite magnetostratigraphy of the Moenave Formation (Fig. 9).

PALEOMAGNETISM, ROCK MAGNETISM, AND MAGNETOSTRATIGRAPHY

The NRM intensities for Moenave Formation strata are typically ~70 mA/m, and more than 90% of the samples have two well-defined components. The first removed component is typically north-directed, with a moderate positive inclination. The first removed component usually unblocks between 300 °C and 600 °C (Figs. 3 and 4) and is interpreted as an overprint magnetization of variable coercivity. These directions do not match expected directions for the

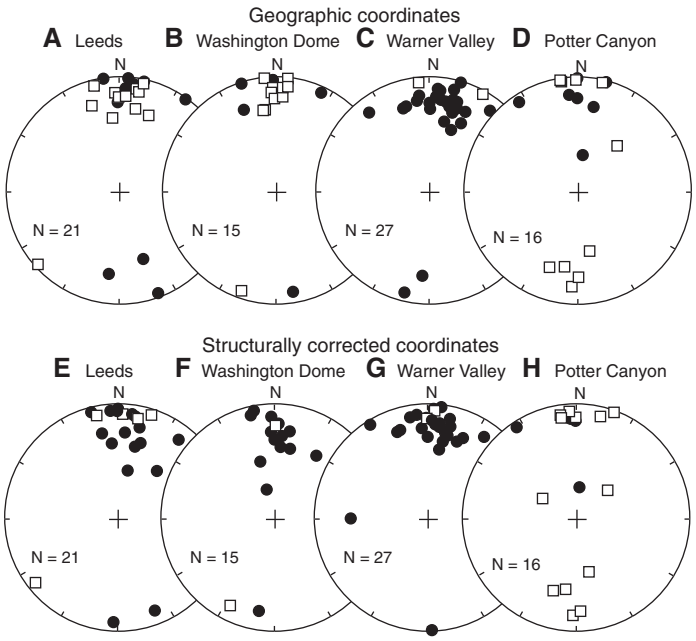


Figure 5. Equal area projections of geographic (A) and tilt-corrected (B) site-mean directions for Leeds, Washington Dome, Warner Valley, and Potter Canyon. Solid circles represent lower-hemisphere projections and open squares represent upper-hemisphere projections.

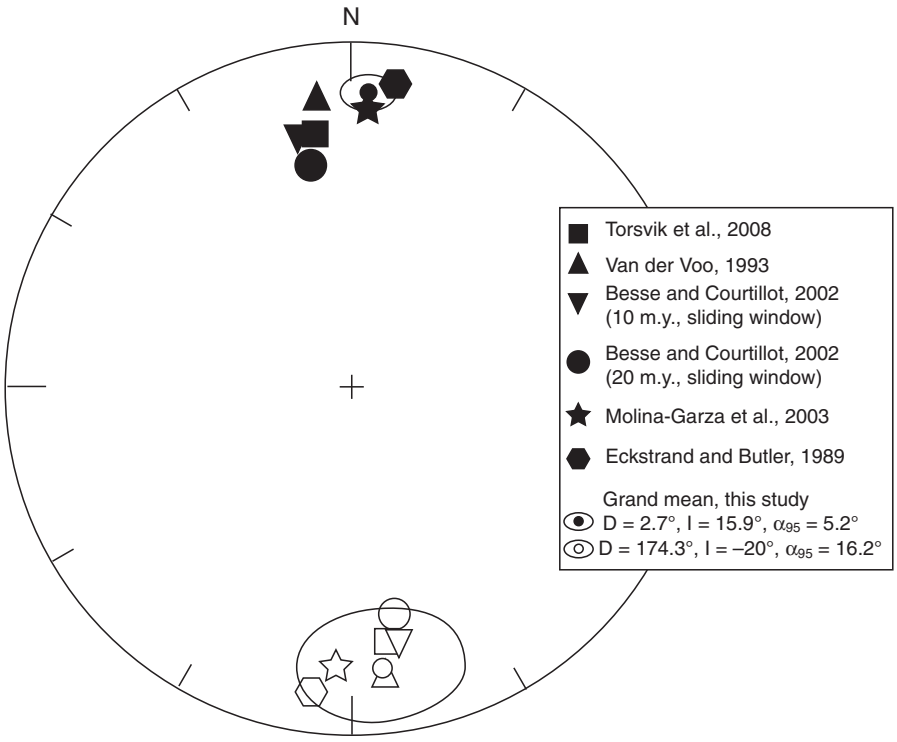


Figure 6. Expected directions calculated from published North American paleomagnetic poles for Triassic–Jurassic aged rocks. Directions were calculated using an average latitude and longitude for the four localities used in this study. The Ekstrand and Butler (1989) and Molina-Garza et al. (2003) poles were derived from results obtained from Moenave strata. The grand mean normal- and reverse-polarity directions calculated in this study are shown with their circles of confidence.

Triassic–Jurassic and are therefore interpreted to be considerably younger. The geologic importance of these overprints is questionable, because paleomagnetic studies of several red beds, of a range of geologic ages, have demonstrated the persistence of secondary magnetizations, of high laboratory unblocking temperatures carried in hematite, that appear to be geologically very young (Dunlop and Stirling, 1977; Ashby et al., 2005; Weil et al., 2006).

The second isolated component, identified in ~87% of the samples, is north directed and shallow (Fig. 3), thus yielding high northern-latitude VGPs. This component is partially isolated by chemical demagnetization and is interpreted as the ChRM (Fig. 3B). In 8% of the samples, the second isolated component yields high southern VGP latitudes and is interpreted as a ChRM of reverse polarity (Fig. 4). In some cases, overprint vector components can make it difficult to isolate the ChRM directions. However, results of remanence stability tests, including reversal and tilt tests, support the conclusion that the ChRM directions are primary. In addition, the overall data set reveals a strong consistency with expected directions for rocks of Late Triassic to Early Jurassic age (Fig. 6) (Ekstrand and Butler, 1989; Van der Voo, 1993; Besse and Courtillot, 2002; Molina-Garza et al., 2003; Torsvik et al., 2008).

Thermal demagnetization of samples treated with three-component IRM reveals a steady decrease in magnetization from 100 °C to ~620 °C. Most of the IRM unblocking is observed in the 3.00 T component as a sharp decrease in magnetization over the laboratory unblocking temperature interval between ~620 °C and 680 °C consistent with fine- to coarse-grained hematite present in these rocks. The 0.30 T component carries ~50%–25% of the total magnetization and also unblocks at ~680 °C. The 0.03 T component is always much less than 25% of the total IRM, and at least 60% of this component is unblocked by 580 °C (Figs. 7A and 7B). Well-indurated, dark-red to purple samples were selected for treatment with chemical demagnetization to assess the degree to which pigmentary hematite could be removed (Collinson, 1967). Samples changed color to white or beige after a cumulative total of ~24 h of soaking in reagent-grade HCl, thus clearly demonstrating the removal of fine-grained, authigenic hematite. These samples were then subjected to progressive acquisition of IRM to a maximum field of 3 T. Saturation IRM intensities range from 3.0A/m to 17A/m, and the IRM acquisition curves reveal a mixture of both low coercivity (possibly magnetite) and higher coercivity (presumably detrital hematite) phases (Figs. 7C and 7D). These samples reach 90%–95% of their total IRM intensity at 1 T and continue to increase in IRM intensity when they reach saturation at ~3 T, a behavior consistent with the presence of detrital hematite. In backfield demagnetization, coercivity of remanence values is ~0.2 T, which is consistent with hematite as the principal magnetic phase. The presence of both authigenic and detrital hematite is supported by petrographic observations (Fig. 8).

TABLE 1. PALEOMAGNETIC DATA FROM THE MOENAVE FORMATION, UTAH AND ARIZONA

Site	Depth (m)	D_{geo} (°)	I_{geo} (°)	D_{tc} (°)	I_{tc} (°)	α_{95} (°)	*k*	n/n_o	VGP Lat (°N)	VGP Lon (°E)
Leeds (37.202°N, 246.650°E)										
N.A.*	69.5		Contact between Whitmore Point Member and Kayenta Formtaion							
184	65.9	35.3	2.5	38.4	13.8	37.3	5.3	4/7	43.9	7.8
157B	63.0	–†	–†	–†	–†	–†	–†	–†	–†	–†
185	61.9	162.3	37.2	157.8	13.0	32.4	6.8	4/6	–41.6	276.7
156B	60.6	–†	–†	–†	–†	–†	–†	–†	–†	–†
154B	58.5	–†	–†	–†	–†	–†	–†	–†	–†	–†
187	52.0	20.2	–30.9	11.7	–12.4	11.2	36.6	6/6	45.2	50
188	52.5	345.6	–5.7	349.6	25.9	11.3	36.1	6/6	64.7	90.8
113	52.0	340.8	–22.3	345.8	5.3	16.7	22.0	5/5	53.1	90.7
181	50.6	21.5	–23.6	6.2	–44.0	11.2	22.1	8/8	60.4	26.8
182	50.0	3.6	0.7	5.1	25.9	52.4	24.9	2/3	66	54.3
183	49.3	228.6	–1.9	231.3	–8.3	27.6	14.1	3/6	–32.7	178.9
N.A.*	49.0		Contact between the Dinosaur Canyon and Whitmore Point Members							
101	48.5	356.4	–15.4	0.6	4.3	8.9	39.8	8/8	54.9	65.6
102	45.6	9.6	–28.1	2.6	–10.2	2.2	62.9	70/70	47.6	62.8
110	40.6	353.0	–36.1	348.1	–9.6	2.9	1031.5	4/5	46.5	84
112	39.3	358.3	–19.1	356.0	7.3	12.5	65.6	3/5	56.3	73.8
111	38.0	2.3	–18.5	359.8	6.5	8.0	92.2	5/5	56.1	67
127B	35.3	10.1	–13.6	10.7	9.7	8.1	153.7	3/3	56.2	47.2
103	34.1	9.8	–12.2	8.9	11.1	6.4	116.6	5/7	57.4	50
109	28.1	11.9	–4.6	13.9	19.9	6.4	143.6	5/5	60.3	38.1
108	21.9	6.6	6.6	13.5	35.9	16.4	32.2	4/4	69.1	28.7
107	15.6	2.8	10.8	17.9	32.4	17.7	19.6	5/5	64.9	23
106	7.6	357.2	22.8	11.6	55.4	6.1	97.6	7/9	80.6	340.8
105	5.3	12.0	1.0	40.1	46	180	1.2	5/5	55	341
105	5.3	188.9	26.9	182.5	8.4	80.3	4.0	2/4	–48.5	242.8
104	3.6	351.7	0.6	355.7	35.4	9.3	42.9	7/8	72	79.8
Locality Mean		1.4	12.0	1.3	13.7	11.1	18.0	11/21		
Washington Dome, (37.009°N, 246.558°E)										
N.A.*	61.0		Contact between Whitmore Point Member and Kayenta Formtaion							
203	59.1	7.0	–0.4	12.7	38.5	5.8	108.6	7/7	71.1	27.6
232	57.3	169.7	10.1	189.6	18	180	1.1	5/5	–42.8	233.6
232	57.3	343.0	1.6	345.8	47.7	6.6	135.8	5/7	75.5	125.8
230	55.7	359.4	2.1	4.7	37.6	9.6	681.9	2/3	73.4	51
205	52.6	–†	–†	–†	–†	–†	–†	–†	–†	–†
229	51.7	198.2	–8.6	206.5	–14.7	44.6	5.8	3/4	–51.9	200.8
228	51.0	–†	–†	–†	–†	–†	–†	–†	–†	–†
227	50.1	–†	–†	–†	–†	–†	–†	–†	–†	–†
225	48.0	–†	–†	–†	–†	–†	–†	–†	–†	–†
208	45.5	347.9	–17.1	33.4	4.7	2.0	27.0	2/4	62.5	61.9
221	43.9	338.7	25.0	344.4	69.0	11.7	75.3	3/6	71.1	216.2
N.A.*	42.5		Contact between the Dinosaur Canyon and Whitmore Point Members							
218	42.0	358.9	–9.5	0.9	31.3	8.8	110.5	4/4	69.8	64.1
210	41.7	353.3	–29.3	346.9	8	11.7	33.7	6/6	54.8	89.7
217	40.9	351.8	–29.7	349.1	4.6	5.1	119.0	4/8	53.8	85.3
216	40.0	357.6	–14.3	358.2	25.9	2.7	598.2	6/6	66.5	71
211	37.4	359.5	–20.5	358.2	11.9	3.8	134.4	12/15	58.9	70.1
239B	36.4	7.4	–26.5	356.8	–9.1	17.5	21.4	4/4	48.3	71.4
212	35.4	3.8	–10.4	5.5	28.4	4.8	449.8	3/3	67.5	52.6
215	32.0	7.0	–8.6	9.8	23.7	10.7	32.5	7/7	63.8	44.5
214	28.0	25.3	5.3	33.9	34.6	9.7	90.0	4/4	55.3	358.6
Locality Mean		359.3	–8.5	1.4	29.7	10.8	18.6	13/15		
Warner Valley, (37.016°N, 246.625°E)										
N.A.*	52.5		Contact between Whitmore Point Member and Kayenta Formtaion							
359B	49.8	14	9.3	24.1	23.7	81.9	3.9	2/4	57.2	19.1
359B	49.8	184.2	25.9	177.4	1.7	96.1	3.1	2/4	–52.1	250.8
356B	46.5	–†	–†	–†	–†	–†	–†	–†	–†	–†

(*continued*)

Each section sampled in the Moenave Formation is dominated by normal polarity (Fig. 9). Notably, thin reverse-polarity magnetozones are observed at each locality. The results from each locality are now described.

Two intervals of reverse polarity are observed at the Leeds locality (Table 1). The oldest interval containing reverse-polarity magnetizations, observed in the Dinosaur Canyon Member (at

TABLE 1. PALEOMAGNETIC DATA FROM THE MOENAVE FORMATION, UTAH AND ARIZONA (*continued*)

Site	Depth (m)	D_{geo} (°)	I_{geo} (°)	D_{tc} (°)	I_{tc} (°)	α_{95} (°)	*k*	n/n_o	VGP Lat (°N)	VGP Lon (°E)
Warner Valley, (37.016°N, 246.625°E) (*continued*)										
N.A.*	44.0		Contact between the Dinosaur Canyon and Whitmore Point Members							
353B	42.5	356.9	36.3	1.2	26.6	38.2	4.5	4/4	67.0	63.3
352B	41.5	24.2	41.3	25.7	29.4	24.4	11.3	4/4	46.4	358.1
328	38.5	322.4	14.1	328.9	4.7	19.7	40.4	3/4	45.1	113.6
326	35.8	354.9	–4.9	359.7	–12.9	16.5	12.2	6/9	46.5	67
325	32.6	15.0	17.8	13	11.5	13.9	19.7	7/8	56.6	42.6
324	32.5	4.9	10.6	3.9	3.7	6.9	135.8	4/4	54.7	59.8
323	29.5	37.6	12.2	35.3	14.5	6.9	76.8	7/7	46.4	10.5
319	28.6	17.1	18.8	14.2	18.1	11.6	34.1	6/6	59.4	38.2
312	21.0	345.0	25.0	343.0	20.5	9.8	33.1	7/7	59.5	101
321	17.5	193.7	9.2	269.1	33.8	180	1.1	3/5	10.3	172.1
320	16.8	352.8	21.4	350.8	11.6	15.1	20.6	5/6	57.7	83.9
311	16.0	7.8	21.3	4.9	19.4	10.1	37.6	6/6	62.6	56.1
310	15.6	18.5	24.3	14.5	25.8	5.1	172.4	6/6	63.2	33.9
318	14.6	12.5	37.5	0.6	29.8	6.9	57.4	7/7	69	65
309	13.6	16.9	29.1	12.0	31.5	5.0	177.0	6/6	67.4	35.4
308	13.5	13.7	24.0	6.9	17.5	6.6	72.0	8/8	61.3	52.3
307	13.2	25.6	34.5	11.1	24.9	9.4	42.4	7/7	64.1	41.2
317	12.3	0.4	24.3	357.5	22.1	10.0	37.5	7/7	64.4	72.3
316	11.8	10.7	21.4	7.7	20.1	11.1	37.5	6/6	62.5	50
315	11.5	25.4	24.5	20.9	27.6	9.3	98.3	4/4	60.9	21.3
314	10.5	6.6	12.5	6.8	2.5	6.9	96.0	6/6	53.7	55.1
313	10.2	16.7	0.7	14.8	–1.4	5.5	276.4	4/7	49.9	43.2
306	6.0	3.9	18.5	1.7	14.7	13.7	24.7	6/6	60.4	63.2
305	5.6	343.0	25.8	341.1	21.4	6.9	124.2	5/5	59	104.7
304	5.5	354.8	17.4	354.6	9.4	47.9	7.7	3/4	57.4	76.6
302	1.6	29.3	–2.9	4.1	–5.4	89.2	10.1	2/9	50.1	60.2
Locality Mean		11.2	21.8	7.3	18.4	6.5	29.2	18/27		
Potter Canyon, (36.879°N, 247.156°E)										
N.A.*	44.3		Contact between Whitmore Point Member and Kayenta Formtaion							
413	43.7	182.7	–16.8	182.3	–16.0	5.6	327.7	3/4	–61.2	242.5
414	42.0	189.0	–32.7	187.7	–37.0	60.8	9.5	2/4	–72.4	222.7
415	40.1	–†	–†	–†	–†	–†	–†	–†	–†	–†
416	36.9	199.3	–29.7	197.6	–34.0	29.7	12.2	3/4	–66.1	202.2
417	34.7	167.1	–45.0	167.0	–50.0	90.0	1.8	2/4	–77.6	311.4
419	30.3	0.3	19.0	360	14	5.6	68.0	11/11	60.2	67.2
420	29.5	355.7	17.0	357.9	13.4	8	71.3	5/5	59.8	71.4
425B	28.8	–†	–†	–†	–†	–†	–†	–†	–†	–†
421	26.3	352.1	–3.6	351.1	–7.4	5	147.8	7/7	48.5	80.7
422	25.7	327.0	8.5	327.2	4.1	8.1	154.2	3/4	43.9	115.9
423	23.4	11.8	26.3	301	–61.5	180	0.7	2/4	–6	106.5
424	18.4	0.7	0.2	356.4	–6	4.6	88.7	8/8	50	72.8
425	17.8	359.8	–2.1	359.9	–11.5	1.9	1722.6	4/4	47.3	67.3
426	15.9	351.3	–1.7	351.3	–11.2	9.2	76.2	4/4	46.7	79.9
427B	11.9	191.6	–28.7	189.8	–23.9	10.4	93.0	3/3	–64.1	224.8
427	11.7	15.4	1.1	19.4	–0.1	11.2	47.9	5/5	48.9	36.8
N.A.*	11.5		Contact between the Dinosaur Canyon and Whitmore Point Members							
428	10.3	10.1	–1.0	10.3	–5.1	7.0	48.3	9/9	49.4	51.2
405	4.6	10.2	64.4	6.0	67.6	24.6	8.5	5/8	75.7	262.9
Locality Mean		359.2	3.7	358.8	2.5	9.2	23.8	13/16		

Note: Samples were collected in southwestern Utah and north-central Arizona. Paleomagnetic data obtained from the Moenave Formation. The first two columns of this table show the site name and stratigraphic position for each locality. The next four columns show the degrees of declination (D) and inclination (I) for geographic (geo) and stratigraphic (tc) coordinates, respectively. The next two columns show the cone of confidence (α_{95}) and precision parameter (*k*). The next column shows the ratio of number of specimens accepted (n) to number of specimens demagnetized for each site (n_o). The last two columns provide the virtual geomagnetic pole position (VGP) calculated for each site given in degrees of latitude (North) and longitude (East of Greenwich).

*N.A. No sample was collected at the boundary.

†–samples are remagnetized.

~5.3 m depth at site 105), yields dual (or mixed) polarity data. Site 105 is composed of nine independent samples. Seven specimens yield well-behaved (principal component analysis results with α_{95} less than 10°), north-directed and shallow demagnetization vectors providing high north-latitude VGPs; two specimens from this site yield south-directed and shallow-inclination magnetizations consistent with high south-latitude VGPs. This site is represented as a half bar on the magnetostratigraphy of the Leeds locality (Fig. 9) because of its mixed polarity character. Site 105 is the only interval of reverse polarity that is observed in the basal part of the Dinosaur Canyon Member at any locality studied here.

A second reverse-polarity interval is observed at ~62 m from the base of the Leeds locality in Whitmore Point strata sampled by site 185. Site 185 is composed of six independent samples. Four samples yield well-behaved, south-directed and shallow-inclination magnetization vectors providing high south-latitude VGPs, and two samples were rejected because the demagnetization vectors are poorly defined. Three sites (154B, 58.5 m; 156B, 60.6 m; and 157B, 63 m) were rejected from the Leeds locality because specimens yield poorly defined demagnetization trajectories, and the poorly defined magnetization components are not consistent with expected directions (Table 1).

At the Washington Dome locality, two intervals exhibit reverse-polarity magnetizations—sites 229, at 51.7 m, and 232, at 57.3 m (Table 1). Site 229 is composed of six independent samples. Four samples record south-directed and shallow-inclination magnetizations consistent with high south-latitude VGPs, and two samples were rejected because demagnetization vectors are poorly defined. Site 232 is composed of 16 independent samples. Five samples display well-behaved, north-directed and shallow-inclination magnetizations, consistent with high north-latitude VGPs. Nine samples record well-behaved, south-directed and shallow-inclination magnetizations consistent with high south-latitude VGPs, and two samples were rejected because the demagnetization vectors are poorly defined. Because site 232 records mixed polarity directions, it is represented as a half bar in the magnetostratigraphy of the Washington Dome locality (Fig. 9). Four sites (225, at 48 m; 227, at 50.1 m; 228, at 51 m; and 205, at 52.6 m) were rejected from the Washington Dome locality because specimens yield poorly defined demagnetization trajectories, and the poorly defined magnetization components are not consistent with expected directions.

At the Warner Valley locality, only one interval (site 359, at ~50 m in the Whitmore Point Member) contains magnetizations of reverse polarity (Table 1). At site 359, eight independent samples were collected. Specimens from two samples display well-behaved, north-directed and shallow-inclination magnetization vectors and thus yield high north-latitude VGPs. Specimens from two samples display well-behaved, south-directed and shallow inclination vector directions and thus high south-latitude VGPs. Specimens from four samples yield poorly defined vector components (Fig. 9). One site, 356B (46.5 m), was rejected from the Warner Valley locality because samples yield poorly defined demagnetization trajectories, and the poorly defined magnetization components are not consistent with expected directions.

At the Potter Canyon locality, one interval of reverse polarity is identified in the lowermost

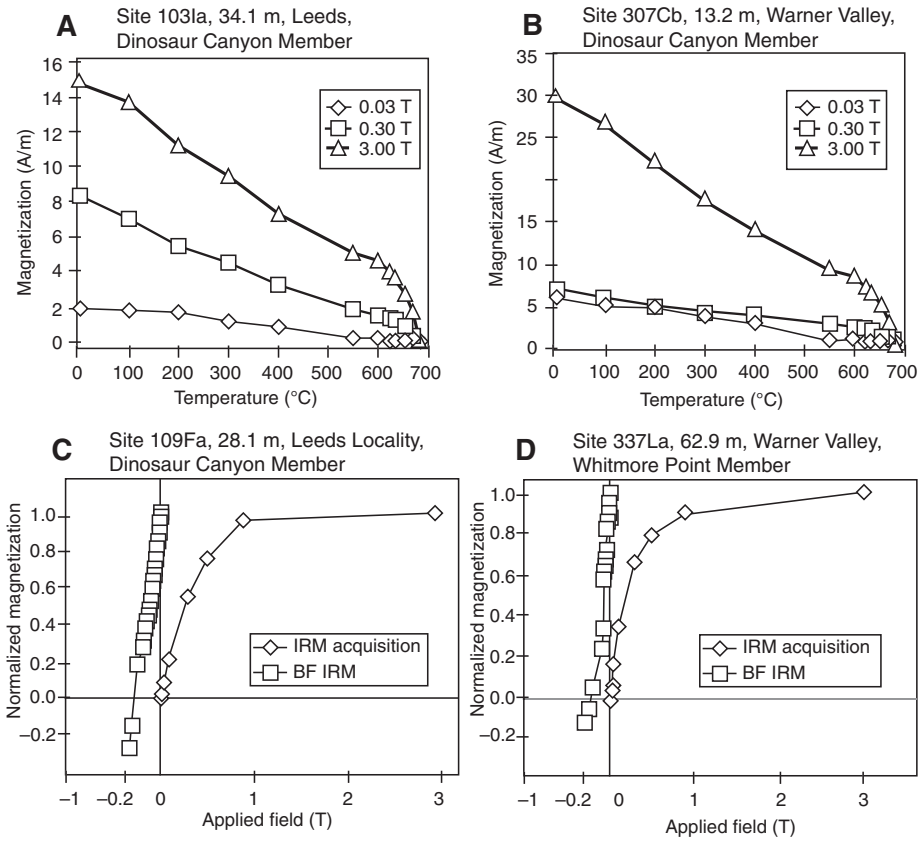


Figure 7. Rock magnetic results from selected sample of the Moenave Formation. Top: examples of three component isothermal remanence magnetization (IRM) thermal demagnetization curves for specimens from the (A) Leeds and (B) Warner Valley localities. Bottom: IRM acquisition and backfield (BF) demagnetization response by samples treated with chemical demagnetization; samples from the (C) Leeds and (D) Warner Valley localities.

Whitmore Point Member at ~11.9 m (site 427B), and one ~9-m-thick interval of reverse polarity is identified in the upper Whitmore Point Member between ~35 and 44 m (sites 417, 416, 414, and 413) (Table 1). Site 427 is composed of seven independent samples, with specimens from each yielding well-behaved, south-directed and shallow-inclination magnetizations providing high south-latitude VGPs. Sites 417, 416, 414, and 413 each consist of four independent block samples, specimens from each yield well-behaved, south-directed and shallow-inclination magnetizations providing high south-latitude VGPs (Fig. 9). Two sites (425B, 28.8 m, and 415, 40.1 m) were rejected from the Potter Canyon locality because samples yield poorly defined demagnetization trajectories, and the poorly defined magnetization components are not consistent with expected directions.

Our collection of paleomagnetic data from the four localities in Moenave strata results in a total of 56 accepted site-mean directions that are less than two sigma away from the grand mean. After application of structural corrections to the data from all four localities, the estimated mean direction for normal-polarity sites (D = 3.4°, I = 7.9°, α_{95} = 2, k = 12.6, n = 53 sites) and the estimated mean for reverse-polarity sites (D = 185.8°, I = –13.2°, α_{95} = 10.9, k = 5.9, n = 3 sites) (Fig. 6) pass an "A" class (γ_c is 5.4° at the 95% confidence) reversal test of McElhinny and McFadden (1990), after inversion of opposite polarity data to common polarity. The mean directions, after inversion to common polarity, for each locality (Table 1) were analyzed using the direction-correction tilt test of Enkin (2003). This analysis resulted in a positive tilt test indicating that the magnetization was probably acquired prior to tilting (72% unfolding at the 95% confidence interval with a standard error of 16.9). The robustness of this test is compromised given that only four locality mean directions could be used for the analysis.

We constructed a composite magnetostratigraphy by correlating reverse-polarity magnetozones defined in the Whitmore Point Member in all four sections and averaging the thickness of the Whitmore Point and Dinosaur Canyon members. The resulting composite section is ~70 m thick. We interpreted the short interval of reverse polarity recorded in Whitmore Point sediments collected at sites 229 of Washington Dome locality and 427 from Potter Canyon locality as representing time-correlative strata. We also interpreted the younger interval of reverse polarity observed in strata from the middle to upper Whitmore Point member including site 185 of the Leeds locality, site 232 of the Washington Dome locality, 359B of the Warner Valley locality, and sites 417–413 of the Potter Canyon locality as representing time-correlative strata. Next, we averaged the thickness of the Dinosaur Canyon member at the Leeds, Washington Dome, and Warner Valley localities to obtain an average thickness for the Dinosaur Canyon Member. Strata of the Dinosaur Canyon Member at the Potter Canyon locality were omitted from this calculation because only the upper part of this member is exposed. We then averaged the thickness of the Whitmore Point Member from all four localities to obtain an average thickness for the Whitmore Point Member. Each interval of reverse polarity was also averaged over all four localities to obtain an average thickness for each magnetozone. The resulting composite magnetostratigraphic section represents an idealized composite in which members may be thicker or thinner than at any specific locality (Fig. 9).

The constructed section, although not necessarily consistent with the specific stratigraphic thicknesses of members at any of the four localities, offers a straightforward way to compare the magnetostratigraphic record of the northern outcrop belt of the Moenave Formation with magnetostratigraphic sections in the UK, Turkey, Italy, Morocco, and eastern North America. The composite magnetostratigraphy shows that most of the Dinosaur Canyon Member of the Moenave Formation is of normal polarity. The composite record includes an ~0.5-m-thick reverse-polarity magnetozone (M1r) at ~4.5 m above the base of the Dinosaur Canyon Member that, notably, is not well defined. The contact between the Dinosaur Canyon and Whitmore Point members lies at ~45 m, and is within normal magnetozone M2n, which is ~42 m thick. The Whitmore Point Member polarity sequence includes two well-defined, reverse-polarity magnetozones, M2r (~1 m thick) and M3r (~5 m thick) (Fig. 9).

DISCUSSION

The refined paleomagnetic and magnetostratigraphic record for the Moenave Formation allows improved correlation among strata

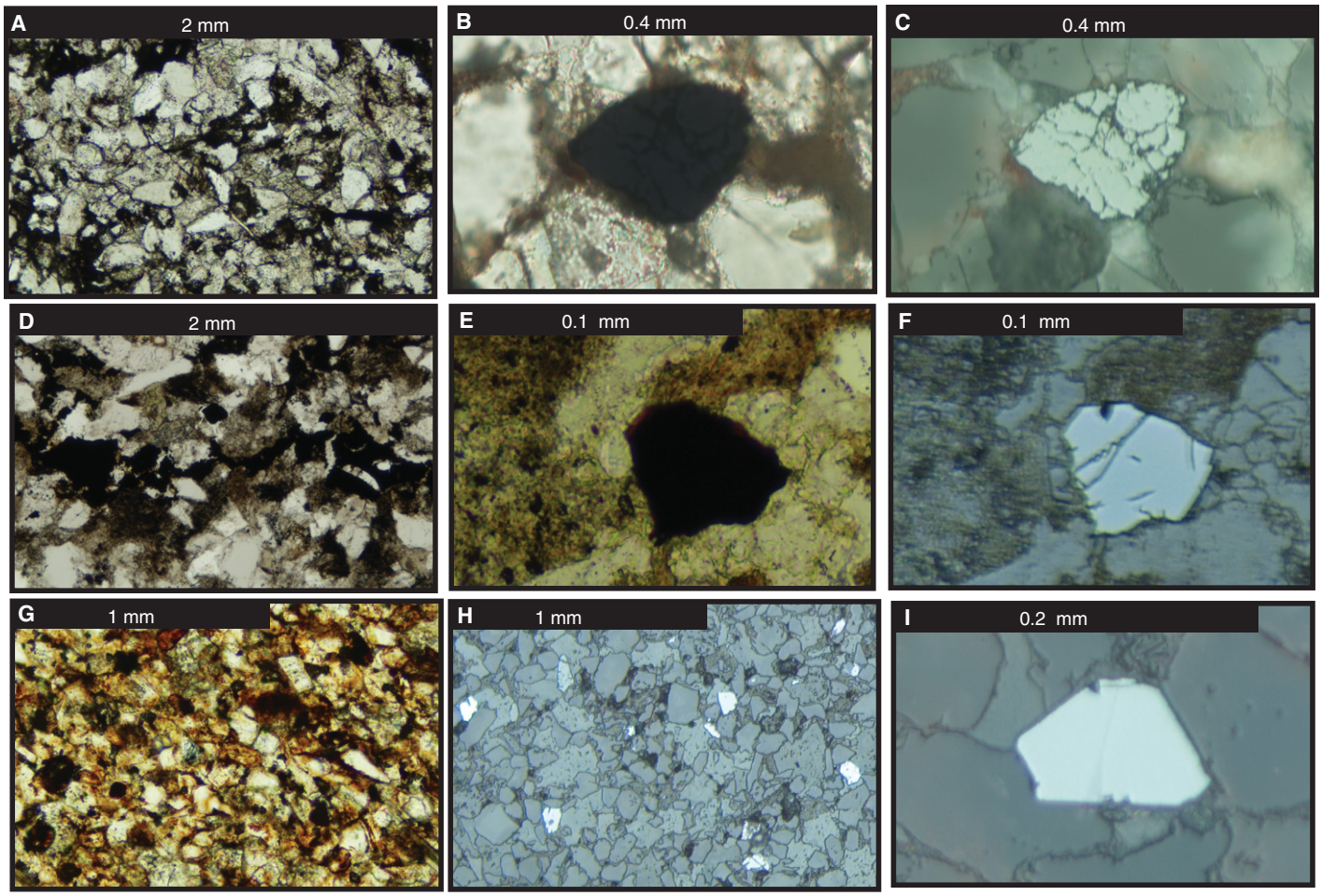


Figure 8. Transmitted- and reflected-light photomicrographs of polished thin sections from representative samples of Moenave strata. Photomicrographs, organized by sample number, show parts of the sample that have been chemically leached (A–D) and parts of fresh samples (E–I). Images of detrital grains are present in all images. Authigenic hematite is observed as red pigment in Figures (E) and (G). Hematite appears as black grains in transmitted light (A, B, D, E, and G) and white grains in reflected light (C, F, H, and I). Samples (A–C) were collected from the Whitmore Point Member, Washington Dome locality, site 232 at ~57.3 m. Samples D–F were collected from the Dinosaur Canyon Member, Leeds locality, site 110 at ~40.6 m. Samples G–I were collected from the Dinosaur Canyon Member, Leeds locality, site 109 at ~28.1 m.

from the American Southwest, Saint Audrie's Bay, UK, Oyuklu, Turkey, Southern Alps, Italy, High Atlas, Morocco, and Newark Basin, eastern North America. Demagnetization results from the four sections studied here show that the Moenave Formation is dominated by normal-polarity magnetozones, and this is consistent with earlier studies of the Moenave Formation as well as our general understanding of the polarity record across the T-J boundary. Accepted site-mean directions of shallow inclination are consistent with an equatorial setting for the American Southwest across the T-J boundary. The correlation of the Moenave record to records of England, Turkey, Italy, Morocco, and eastern North America are now described.

We correlated intervals of reverse polarity in strata of the Whitmore Point Member to facilitate combining the magnetostratigraphic records from our four localities. We also used an average member thickness to obtain a composite magnetostratigraphy for the Moenave Formation (Fig. 9). The composite magnetostratigraphy shows that the Moenave Formation was deposited during mostly normal polarity (magnetozones M1n, M2n, M3n, and M4n). The oldest interval of reverse polarity, M1r, is identified in Dinosaur Canyon strata by part of site 105, at 5.3 m, at the Leeds locality, although the data from this thin interval, as described above, are less than robust. The subsequent interval of reverse polarity, M2r, is defined in Whitmore Point strata by sites 229, at 51.7 m, at the Washington Dome locality, and 427B, at 11.9 m, at the Potter Canyon locality. The youngest interval of reverse polarity, M3r, is defined in Whitmore Point strata by sites 185, at 61.9 m, at the Leeds locality, part of site 232, at 57.3 m, at the Washington Dome locality, part of site 359B, 49.8 m, of the Warner locality, and 417, 416, 414, and 413, at 34.7 m to 43.7 m, at the Potter Canyon locality.

Rock magnetic data show that the principal remanence carrier in the Moenave sediments is hematite. Three-component (0.03 T, 0.3 T, 3 T) thermal demagnetization experiments show that the IRM is of high coercivity and fully unblocks at 680 °C, the signature unblocking temperature for hematite. Data obtained from chemical leaching of pigment hematite fol-

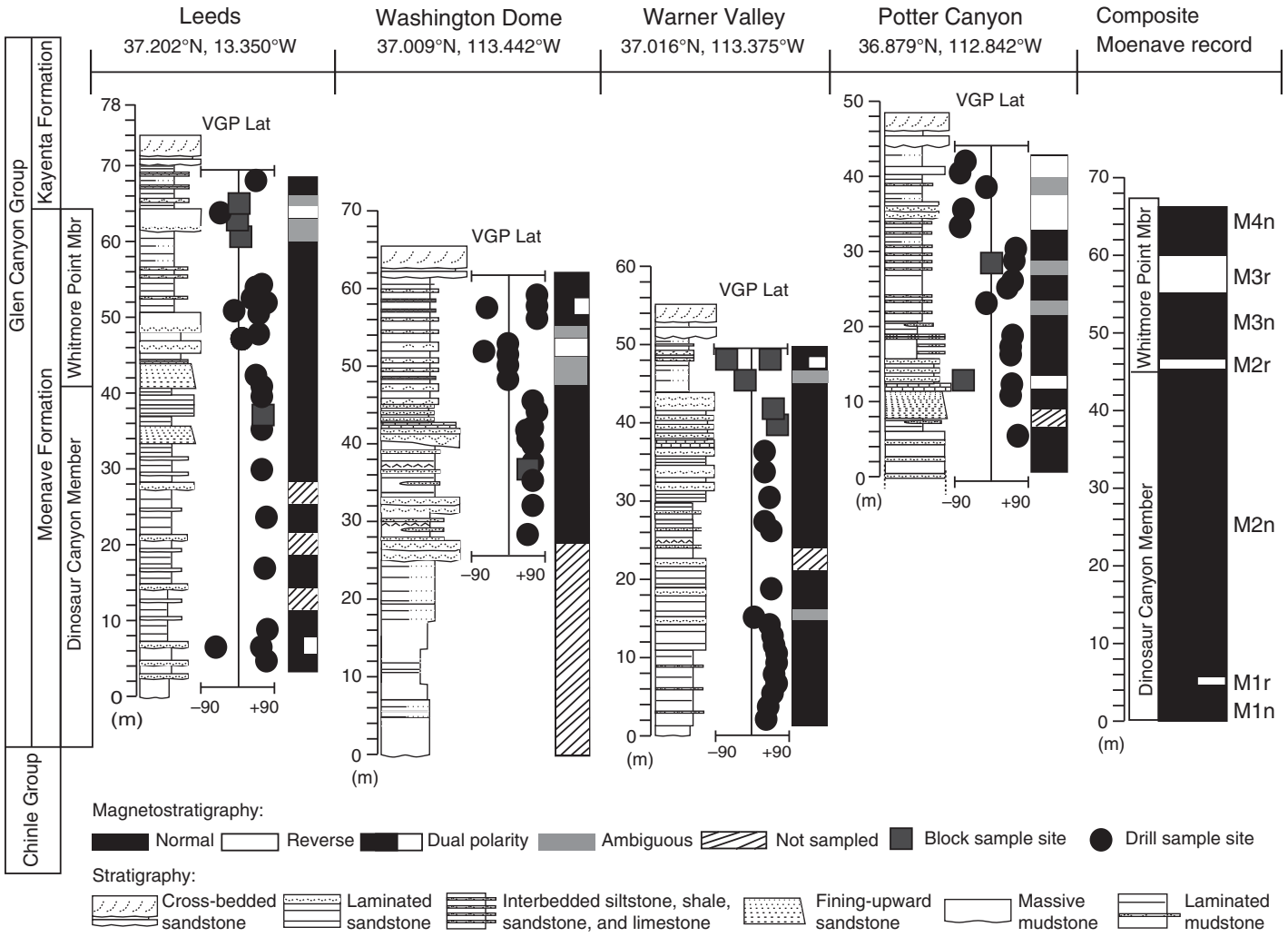


Figure 9. Stratigraphic logs, virtual geomagnetic pole (VGP) latitudes based on site-mean directions, and magnetostratigraphic records for the Leeds, Washington Dome, Warner Valley, and Potter Canyon localities, along with a composite Moenave Formation record. Intervals that could not be sampled that are thicker than 5 m are identified in the hashed pattern.

lowed by saturation IRM acquisition show that detrital hematite constitutes at least part of the hematite-dominated remanence. Petrographic observations show that authigenic (translucent or pigment) hematite is ubiquitous, although irregularly distributed. Hematite of detrital origin, as coarser opaque grains, is very rare, and is less than 0.1% by volume of detrital grains. Detrital magnetite has not been identified by petrographic inspection. Thin sections prepared from specimens that have been subjected to progressive chemical demagnetization in reagent HCl show little if any remaining authigenic hematite (Fig. 8).

The ChRM directions from the four Moenave Formation sections, as well as the previous studies by Ekstrand and Butler (1989), are consistent, at least in inclination, with expected directions for Triassic–Jurassic rocks based on several compilations (Fig. 6). We interpret the ChRM in the Moenave Formation to be primary, acquired very close to the time of deposition. Normal- and reverse-polarity magnetizations yield a positive reversal test, although the test is not as robust as possible due to dominance by normal polarity. A positive tilt test supports the hypothesis that the ChRM was acquired prior to tilting. We interpret this correspondence to suggest that the remanence acquisition process in most Moenave strata was of sufficient fidelity to record the geomagnetic field, including short polarity chrons, over the Triassic–Jurassic boundary interval. Remanence acquisition occurred over a sufficiently short period of time to provide an internally consistent polarity record (Beck et al., 2003). Possible heterogeneities in the magnetization acquisition process in the oldest part of the Dinosaur Canyon Member in the northern outcrop belt resulted in a slower rate of remanence acquisition and are reflected in intervals of mixed remanence polarity, which are of lesser quality in comparison to the younger reverse magnetozones. Correlations based on the M1r magnetozone are not robust and should be further examined in other Dinosaur Canyon sections.

Biostratigraphic evidence, including conchostracans, pollen, and *Eubrontes* tracks, all indicate a latest Triassic age for the lowermost part of the Whitmore Point Member (Kirkland et al., 2002; Cornet and Waanders, 2006; Milner et al., 2006; Kozur and Weems, 2010). However, upper Dinosaur Canyon strata in the southern outcrop belt contain Jurassic *Protosuchus* remains. These strata are considered to be laterally equivalent to upper Whitmore Point strata in the northern outcrop belt (Lucas and Heckert,

2001; Lucas and Tanner, 2007a). The exact relationship between the fossil-bearing strata in the southern outcrop belt and the reverse-polarity magnetozones that we have defined from the Whitmore Point Member in the northern outcrop belt is uncertain, because there are no magnetostratigraphic data from the locality where *Protosuchus* was identified. We infer that the fossil-bearing strata in the southern outcrop belt are younger than the youngest stratigraphic interval of reverse polarity in the northern outcrop belt, because lowermost Jurassic strata are of normal polarity (Lucas and Heckert, 2001; Molina-Garza et al., 2003). If our interpretation is correct, then both magnetozones M2r and M3r are of latest Triassic age with M3r likely to be during the Triassic-Jurassic boundary interval (Fig. 10).

In the previous paleomagnetic studies of the Moenave Formation, one reverse-polarity magnetozone was reported from both the northern (Vermillion Cliffs: Ekstrand and Butler, 1989) and the southern (Echo Cliffs: Molina-Garza et al., 2003) outcrop belts in Moenave Formation strata sampled close to the sub-Springdale unconformity (Fig. 2). Given the lateral extent of exposures of the Whitmore Point Member of the Moenave Formation and the proximity of both reverse magnetozones to the sub-Springdale unconformity, Molina-Garza et al. (2003) argued that these were the same reverse magnetozones and inferred a latest Triassic age for strata below the Springdale unconformity, because they correlated the reverse-polarity magnetozone with reverse-polarity magnetozone E23r of the Newark polarity record. Our new composite magnetostratigraphy for the Moenave Formation shows at least two, well-defined, reverse-polarity magnetozones in uppermost Triassic strata. We support the hypothesis that the previously recognized reverse-polarity magnetozones stratigraphically high in Moenave strata in the northern and southern outcrop belts of the Moenave Formation are equivalent and may correlate to either the M2r or M3r magnetozones. Further magnetostratigraphic work is needed in the southern outcrop belt of the Moenave Formation to clarify the timing of deposition of strata between the northern and southern outcrop belts.

A correlation between the Moenave Formation and the stratigraphic succession at Saint Audrie's Bay, UK (Hounslow et al., 2004) is critical for comparing the relative timing of T-J boundary events recorded in both marine and nonmarine strata. We infer that the interval of time represented by reverse-polarity magnetozone M1r of the Moenave Formation correlates with the interval of time represented by SA5n.1r of the Saint Audrie's Bay Record (Fig. 10). Notably, magnetozone M1r is not as well defined as younger reverse-polarity magnetozones in the Moenave Formation, so this correlation is tentative. The interval of time characterized by reverse-polarity magnetozone M2r of the Moenave Formation must correlate to either the interval of time represented by reverse magnetozone SA5n.2r or SA5n.3r of the Saint Audrie's Bay record, because both magnetozones lie in strata of demonstrably Triassic age. We infer that the interval of time characterized by reverse-polarity magnetozone M3r correlates with magnetozone SA5n.5r, which occurs within the 6-m-thick interval in the Saint Audrie's Bay section that encompasses the T-J

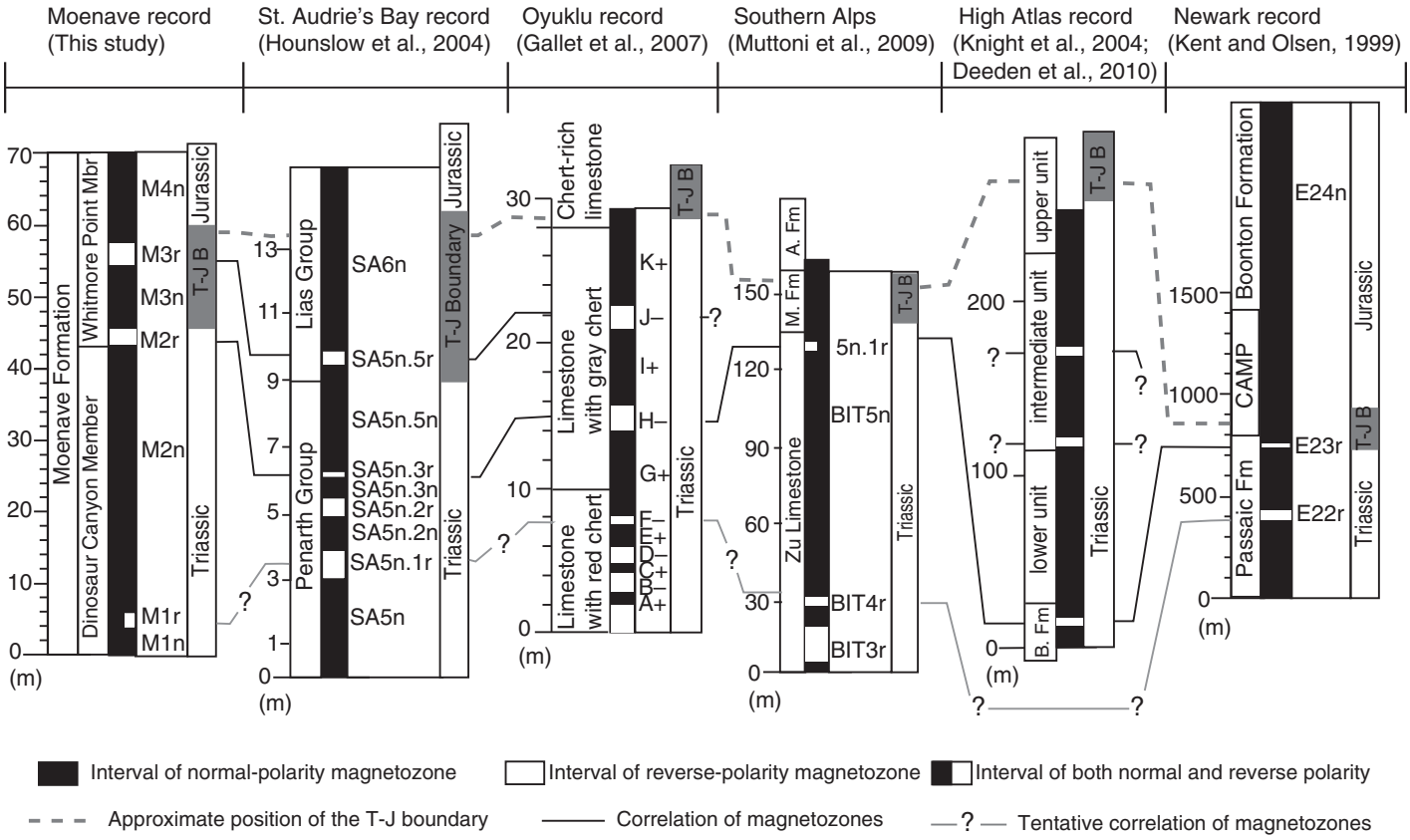


Figure 10. Correlation of the composite Moenave Formation magnetostratigraphy with magnetostratigraphic records from the marine succession at Saint Audrie's Bay, UK, Oyuklu, Turkey, Southern Alps (M. Fm.—Malanotte Formation; A. Fm.—Albenza Formation), Italy, and continental sections in High Atlas (B. Fm.—Bigoudine Formation), Morocco, and continental deposits of the Newark and Hartford basins, eastern North America. CAMP—central Atlantic magmatic province.

boundary. The stratigraphic interval between the bottom of M2r and the top of M3r includes ~15 m of Whitmore Point strata. The equivalent stratigraphic interval at Saint Audrie's Bay is an ~7 m interval from the bottom of SA5n.2r to the top of SA5n.5r. The initial carbon isotope shift at Saint Audrie's Bay occurs in the same stratigraphic interval as magnetozone SA5n.4n, and the main carbon isotope shift occurs in strata characterized by magnetozone SA6n, stratigraphically younger than reverse magnetozone SA5.5r (Hesselbo et al., 2002). Again, we tentatively correlate this interval SA5n.4n to SA6n, to the interval in Moenave strata characterized by magnetozones M3n to M4n. Investigation of the isotopic record of the Moenave Formation is necessary to evaluate the validity of our proposed correlation.

Although M1r is less well defined than other reverse-polarity magnetozones, if we assume that M1r of the Moenave Formation represents a short time interval of true reverse polarity, then it likely correlates with reverse-polarity magnetozone F– of the marine Oyuklu section in Turkey. The interval of time spanning reverse-polarity magnetozones M2r and M3r in Triassic strata of the Moenave Formation most likely correlates with latest Triassic magnetozone intervals H– to J– of the Oyuklu section (Fig. 10). The ~15 m of Moenave Formation strata between magnetozones M2r and M3r correlates to ~8 m of section in marine strata from Oyuklu, Turkey. A single reverse-polarity magnetozone (0.1r) has been identified in strata close to the T-J boundary in the Italcementi Quarry section of the Southern Alps. This magnetozone, defined by data from only one sampling interval (Muttoni et al., 2009), may correlate with either M2r or M3r in the Moenave Formation.

We do not find it possible to correlate M1r of the Moenave record with any part of the High Atlas, Morocco, polarity record. Reverse-polarity magnetozone M2r, on the other hand, may best correlate with the reverse-polarity magnetozone within the Bigoudine Formation of the High Atlas section, because both reverse magnetozones lie in Triassic strata (Fig. 10). The interval of time characterized by the younger reverse magnetozone in the "intermediate unit" of the High Atlas section most likely correlates with the interval represented by M3r of the Moenave Formation, reverse magnetozone SA5n.5r of the Saint Audrie's Bay section, and reverse magnetozone J– of the Oyuklu section. This time interval of reverse polarity has not been identified in strata of the Newark Basin. The ~15 m interval of Moenave Formation strata between M2r and M3r is time equivalent to ~100 m of central Atlantic magmatic province extrusive rocks and intercalated continental sediments from the High Atlas record.

We tentatively correlate the M1r oldest reverse-polarity magnetozone in the Moenave Formation to reverse-polarity magnetozone E22r of the Newark record. As with the other correlations using the M1r interval, the correlation is most tentative. We correlate reverse-polarity magnetozone M2r of the Moenave Formation to reverse-polarity magnetozone E23r of the Newark record, which is inferred to have a duration of ~25 ka, corresponding to ~3 m of sedimentation (Kent and Olsen, 1999, 2008). Notably, both reverse-polarity magnetozones lie in strata with similar biostratigraphic characteristics, including the ichnogenus *Eubrontes* and a change in pollen species dominated by *Classopollis meyerianus* followed by a shift to *Classopollis torusus* (Kirkland et al., 2002; Cornet and Waanders, 2006; Milner et al., 2006; Lucas, 2009; Kozur and Weems, 2010). We do not find it possible to correlate M3r of the Moenave record with any part of the Newark polarity record. However, the upper Whitmore Point strata, within normal-polarity magnetozone M4n, contain earliest Jurassic conchostracan fauna characterized by *Bulbilimnadia killianorum*, as do strata of the Feltville Formation, Newark Basin, which are within normal-polarity magnetozone E24n (Kozur and Weems, 2010). We thus speculate that the ~15 m interval of Moenave Formation strata between magnetozones M2r and M3r is time equivalent to ~180 m of Newark strata from the bottom of E23r in the upper Passaic Formation to the basal Feltville Formation, including the Orange Mountain Basalt. It is difficult to compare the thickness of a similar time interval of Moenave strata with rift-basin strata of the Newark Basin because sedimentation rates are so different. This comparison is also hindered by the absence of a latest Triassic reverse-polarity magnetozone in the Newark record that would correlate with M3r of the Moenave Formation.

Our proposed correlations of the Moenave magnetostratigraphic record from the American Southwest with polarity records from the southern UK, southwest Turkey, Southern Alps, Italy, Morocco, Africa, and eastern North America establish a chronology that can be used to interpret the timing of some events that characterize the Triassic–Jurassic boundary. The oldest well-defined geomagnetic polarity feature observed in the Moenave record is the latest Triassic-age, reverse-magnetozone M2r, which we correlate with the reverse magnetozones SA5n.2r or SA5n.3r of Saint Audrie's Bay, UK, magnetozone H– of the Oyuklu record, Turkey, 0.1r in the Italcementi Quarry record in the Southern Alps, Italy, the reverse-polarity magnetozone of the Bigoudine Formation, Morocco, and magnetozone E23r of the Newark Basin. A younger, latest Triassic reverse-polarity magnetozone, M3r, in the Moenave record is correlated with SA5n.5r of Saint Audrie's Bay, magnetozone J– of the Oyuklu record, and the interval of reverse polarity of the "intermediate unit," High Atlas, Morroco (Fig. 10). The geomagnetic field is characterized by normal polarity during earliest Jurassic time when terrestrial biostratigraphic records include *Protosuchus* fossils, *Bulbilimnadia killianorum* conchostracan fauna, and when *P. spelae* and *P. planobis* ammonites appear in marine records.

CONCLUSIONS

The composite magnetostratigraphic record for the Dinosaur Canyon and Whitmore Point members of the Moenave Formation of the American Southwest reported here facilitates correlation with both marine and nonmarine successions deposited across the T-J boundary. These sections include the Penarth and Lias Groups at Saint Audrie's Bay, UK, the limestone with red chert, limestone with gray chert, and chert-rich limestone of the Oyuklu record, Turkey, the middle Zu Limestone to the lower Albenza Formation at the Italcementi Quarry, Southern Alps, Italy, the Bigoudine Formation, "lower unit, intermediate unit, and upper unit" in High Atlas, Morocco, the Passaic Formation of the Newark Basin, and central Atlantic magmatic province eruptions of both Morocco and eastern North America. These correlations aid in evaluating the relative timing of events near the T-J boundary in marine and nonmarine strata. The oldest reverse magnetozone in the Dinosaur Canyon Member of the Moenave Formation (M1r) is defined by mixed polarity results, and this is less robust than younger intervals of reverse polarity in the Moenave Formation. Magnetozone M1r is only used for the most tentative of correlations. Reverse-polarity magnetozone M2r is correlated, on the basis of biostratigraphic information, to either magnetozones SA5n.2r or SA5n.3r of Saint Audrie's Bay, magnetozone H– of Oyuklu, magnetozone 0.1r of Italcementi Quarry, the reverse-polarity magnetozone of the Bigoudine Formation, High Atlas, Morocco, and magnetozone E23r of the Newark Basin. The youngest magnetozone in the Moenave Formation, M3r, from the middle part of the Whitmore Point Member, is of latest Triassic age and correlates to SA5n.5r of Saint Audrie's Bay, J– of the Oyuklu section, and the interval of reverse polarity form the "intermediate unit," Morocco. This reverse-polarity magnetozone must correspond to a time interval

between the emplacement of the Orange Mountain Basalt and early deposition of Feltville Formation strata prior to the lowest occurrence of the Hettangian conchastracan *Bulbilimnadia killianorum*. If our proposed correlations are viable, then the T-J boundary interval likely lies within normal-polarity magnetozones M4n in the Moenave record, SA6n in the Saint Audrie's Bay record, K+ of the Oyuklu record, the normal-polarity interval above 0.1r in the Italian record, the normal magnetozone of the "upper unit" in Morocco, and E24n of the Newark record (Fig. 10).

Most of the Moenave Formation, ~58 m (out of ~70 m), between the base of the Dinosaur Canyon Member and the middle Whitmore Point Member, was deposited in the latest Triassic. The facies change from mostly fluvial Dinosaur Canyon to mostly lacustrine Whitmore Point occurred during the latest Triassic, in response to an inferred drop in base level (Tanner and Lucas, 2009b). The rich fossil assemblages, including but not limited to *Eubrontes* and *Gralator* dinosaur tracks, semionotid fish, plant fossils, and invertebrates, are all preserved at the Dinosaur Discovery Site at Johnson Farm in strata that lie ~4 m above the transition from the Dinosaur Canyon Member to lower Whitmore Point strata. The fossil assemblage preserved at the Dinosaur Discovery Site is therefore of latest Triassic age, rather than Early Jurassic age as previously interpreted by Kirkland et al. (2002), Cornet and Waanders (2006), and Milner et al. (2006). Our new chronostratigraphic interpretation of the events surrounding the T-J boundary interval support the initiation of central Atlantic magmatic province volcanism, carbon isotope fluctuations, terrestrial mass extinction, and subsequent flora and faunal radiation as latest Triassic events.

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